Eyes are smarter than scientists believed

Steeve Laquitaine
& Ulisse Ferrari

laquitainesteeve@gmail.com

INSTITUT DE LA VISION PARIS

Material at **https://tinyurl.com/ensta2025**

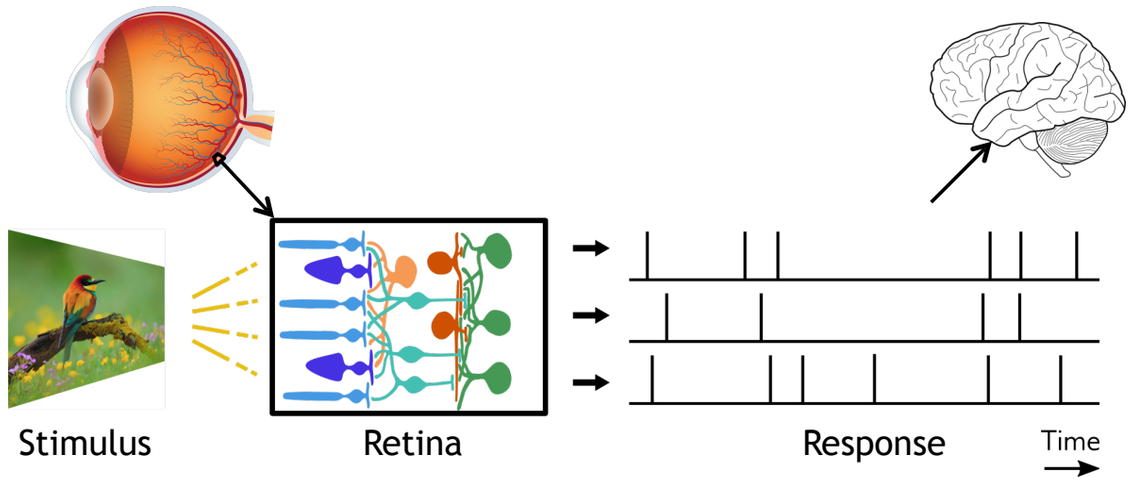
Eyes are smarter than scientists believed

1) The structure of the retina

2) Stimulus processing in the retina

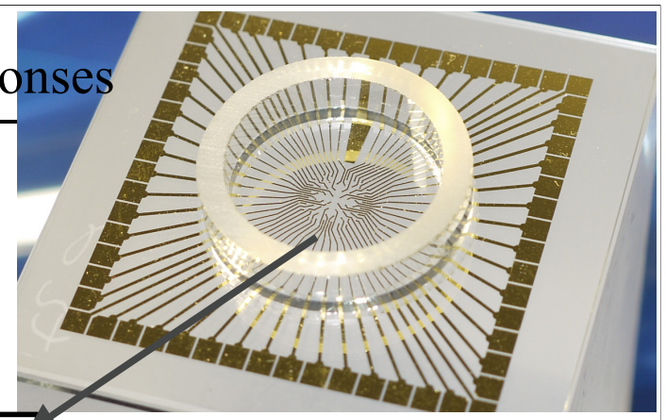
3) Predicting retinal light-response

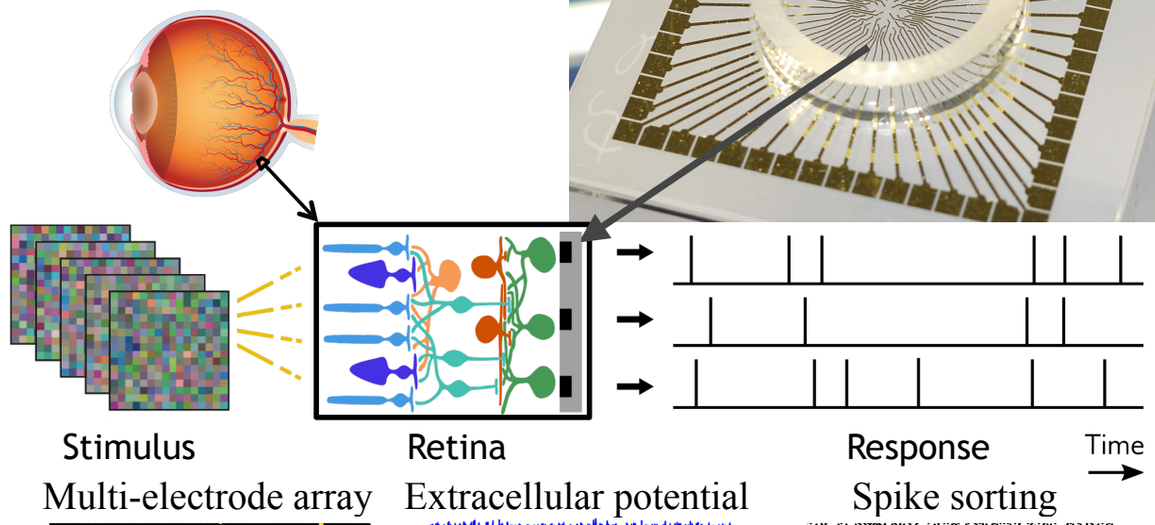
Predicting retinal light responses



Goal: predict spiking times

Predicting retinal light responses

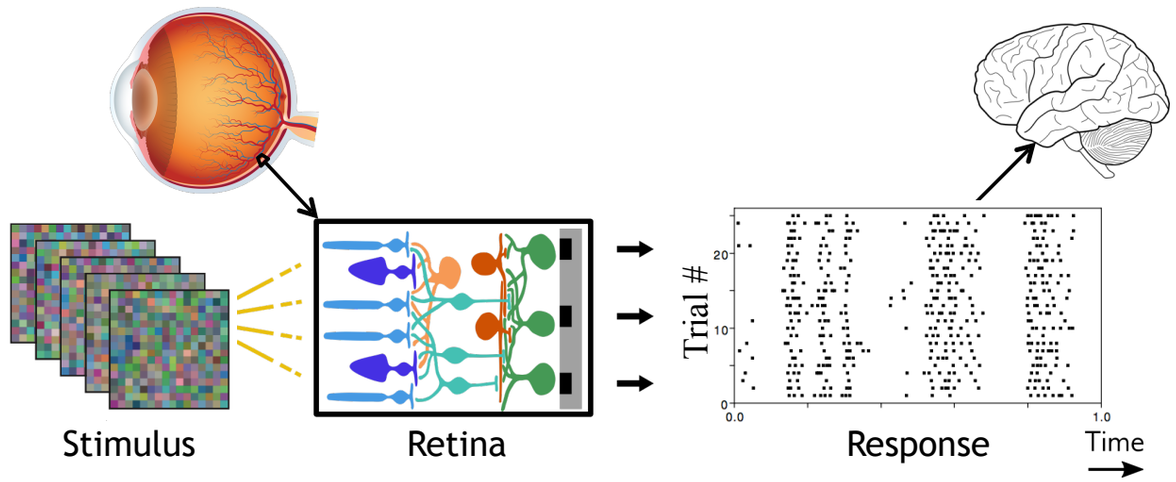




Marre '12; Yger '16

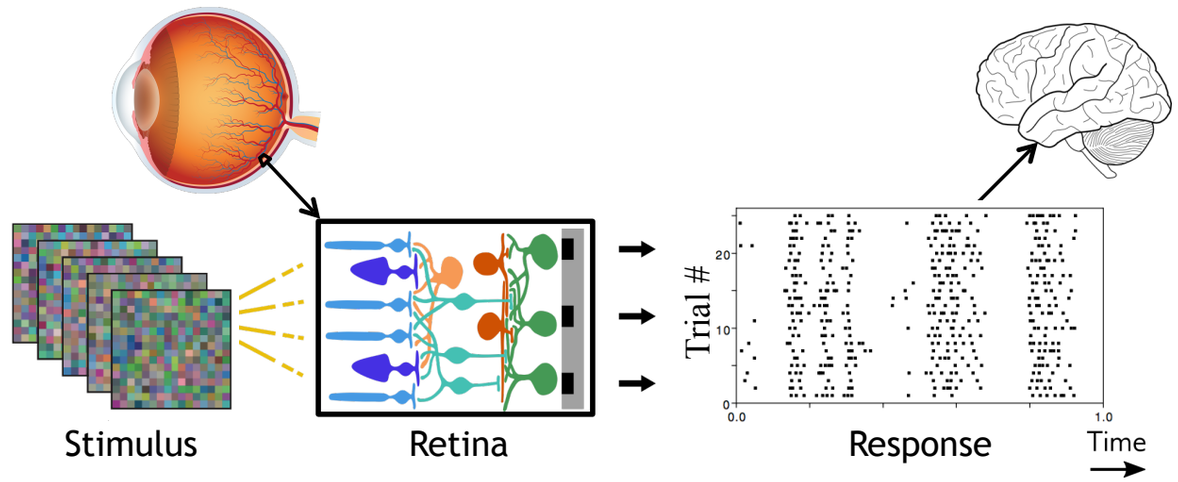
Predicting retinal light responses

Predicting retinal light responses



Response is reliable but noisy!

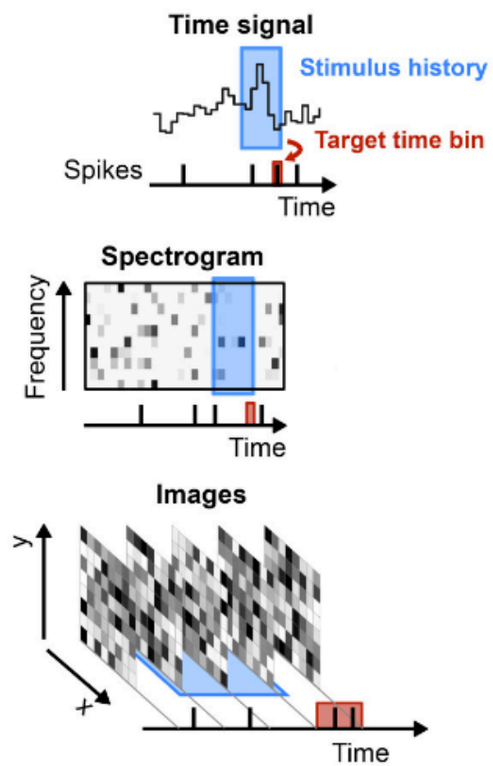
Predicting retinal light responses



Response is reliable but noisy!

Average over trials

Predicting retinal light responses



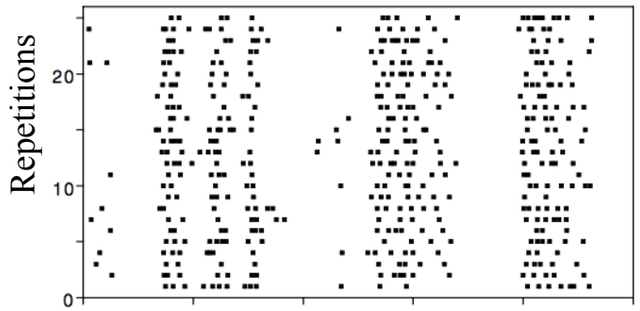
The approach generalizes
to other stimulus domains

Meyer Front. Syst. Neurosci, 2017

Predicting retinal light responses

Peri-Stimulus Time Histogram (PSTH)

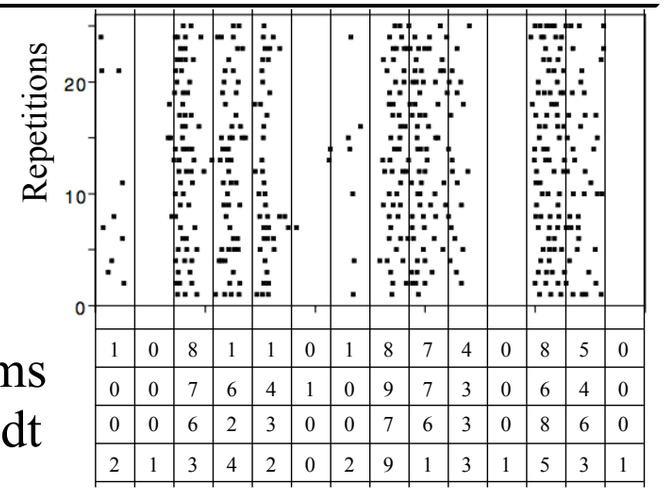
1) repeat video (trials)
2) align responses in time



Predicting retinal light responses

Peri-Stimulus Time Histogram (PSTH)

1) repeat video (trials)
2) align responses in time
3) bin responses at dt=25ms
4) spike counts n_t in each dt

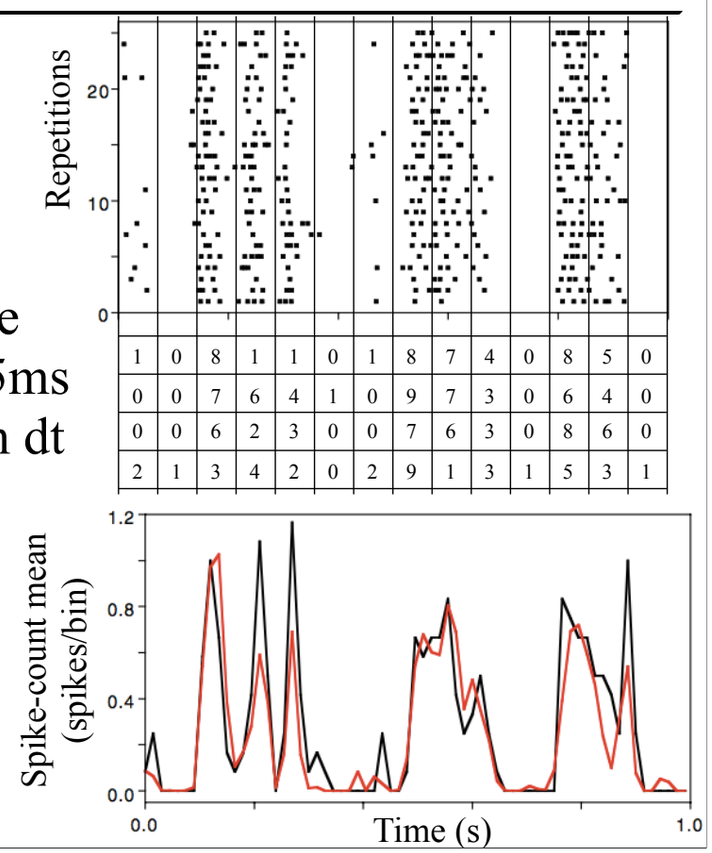


Predicting retinal light responses

Peri-Stimulus Time Histogram (PSTH)

1) repeat video (trials)
2) align responses in time
3) bin responses at dt=25ms
4) spike counts n_t in each dt
5) average over trials
6) estimate spike rate

Can we predict $\langle n_t \rangle$?



Linear-Gaussian model

$$\tilde{S}_{xy}(t) = \Big(S_{xy}(t) - \langle S \rangle \Big) \Big/ \mathrm{std}(S)$$ The stimulus

$$f(t) = w * \tilde{S}(t) + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$ The model

Linear-Gaussian model

$$\tilde{S}_{xy}(t) = \left(\boxed{S_{xy}(t)} - \langle S \rangle \right) \Big/ \mathrm{std}(S)$$

Take stimulus at time t

$$f(t) = w * \tilde{S}(t) + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$

Linear-Gaussian model

$$\tilde{S}_{xy}(t) = \boxed{\Big(S_{xy}(t) - \langle S \rangle\Big) \Big/ \mathrm{std}(S)}$$

Normalize its amplitudes

$$f(t) = w * \tilde{S}(t) + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$

Linear-Gaussian model

$$\tilde{S}_{xy}(t) = \Big(S_{xy}(t) - \langle S \rangle \Big) \Big/ \mathrm{std}(S)$$

$$\boxed{f(t)} = w * \tilde{S}(t) + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$

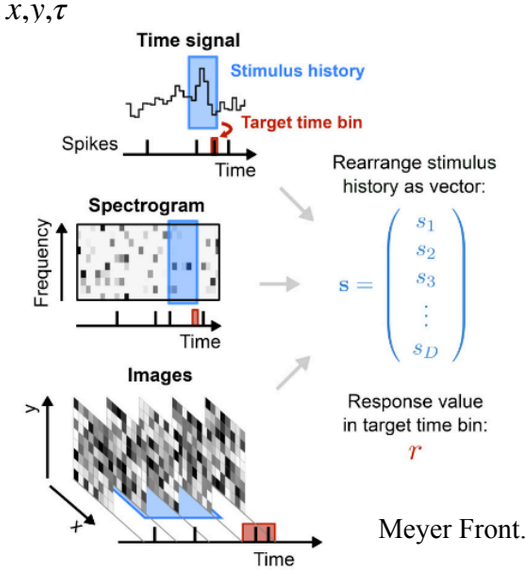
Use a model to predict its neural response

Linear-Gaussian model

$$\tilde{S}_{xy}(t) = \Big(S_{xy}(t) - \langle S \rangle \Big) \Big/ \mathrm{std}(S)$$

$$f(t) = w * \boxed{\tilde{S}(t)} + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$

Make
a vector



Meyer Front. Syst. Neurosci, 2017

Linear-Gaussian model

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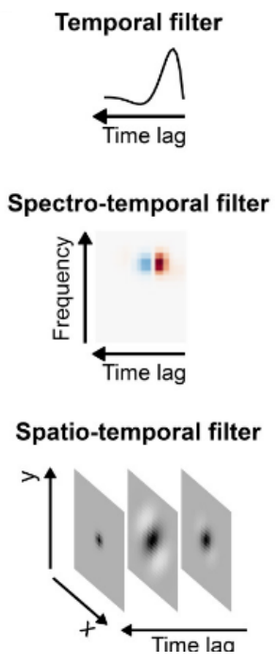
$$f(t) = \boxed{w} * \tilde{S}(t) + b = \sum_{x,y,\tau} w_{x,y}(\tau)\ \tilde{S}_{x,y}(t-\tau) + b$$

Learn a filter **w**



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Project stimulus onto filter

Linear-Gaussian model

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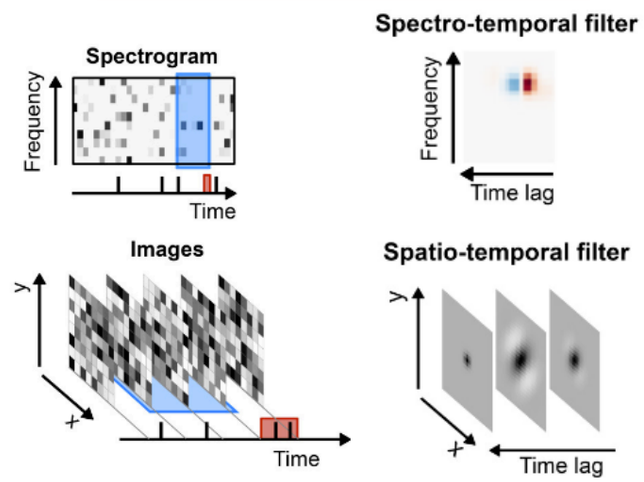

Integrate over **space and time**

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Linear-Gaussian model

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How do we learn **w?**

Linear-Gaussian model

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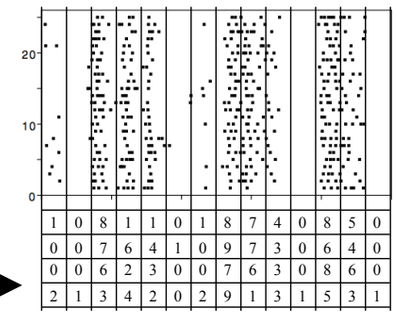
How do we learn **w?**

We have the actual spike rate n(t)



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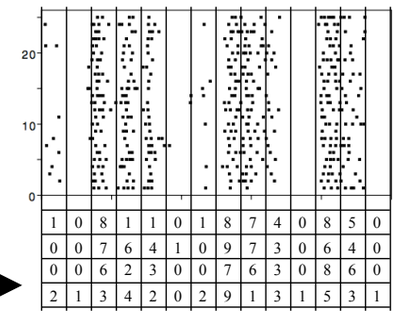

We have the predicted rate f(t)

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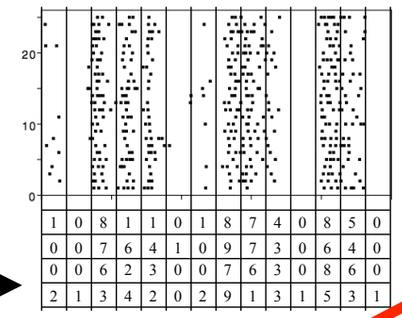
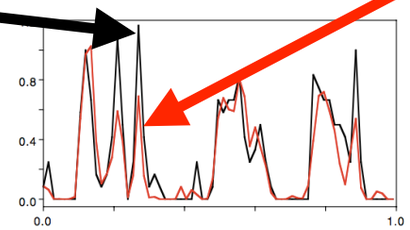


We have the predicted rate f(t)

looks like this …



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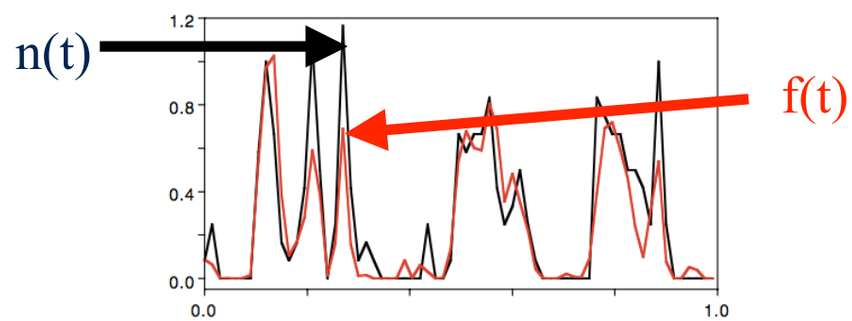
Optimise w by minimising: $\mathrm{MSE} = \dfrac{1}{2T} \sum_{t} \big(f(t) - n(t)\big)^2$



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After some algebra:

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$$b = \langle n \rangle$$

After some algebra:

$$\boxed{w_{xy}(\tau)} = \sum_{x'y'\tau'} C^{-1}_{xy,x'y'}(\tau,\tau')\ \mathrm{STA}_{x'y'}(\tau')$$

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Spike-triggered average

$$\mathrm{STA}_{xy}(\tau) \equiv \frac{1}{T} \sum_t^T n(t)\ \tilde{S}_{xy}(t-\tau)$$

Linear-Gaussian model

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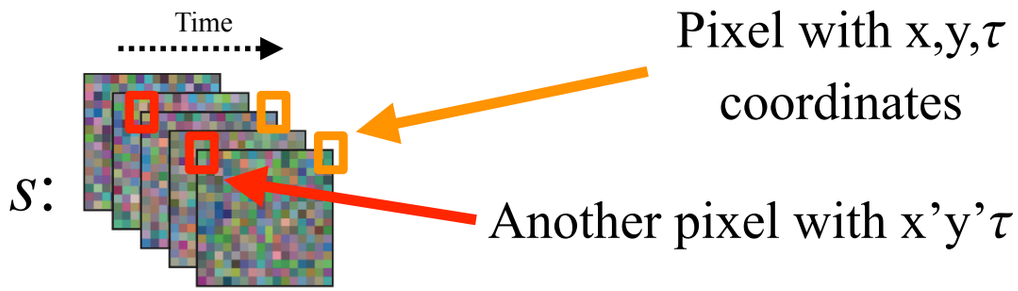
Stimulus autocorrelation

$$C_{xy,x'y'}(\tau',\tau) \equiv \frac{1}{T} \sum_t^T \tilde{S}_{xy}(t-\tau')\ \tilde{S}_{x'y'}(t-\tau)$$

Linear-Gaussian model - Terminology



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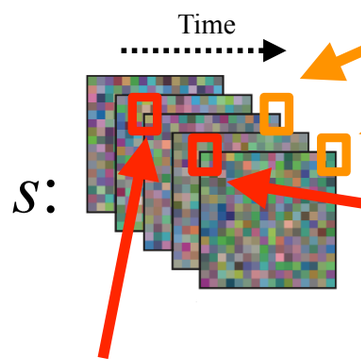

Pixel with x,y,τ'

Pixel with x,y,τ coordinates

Another pixel with x'y'τ

pixel at x'y',τ'

$$w_{xy}(\tau) = \sum_{x'y'\tau'} \boxed{C^{-1}_{xy,x'y'}(\tau,\tau')}\, \mathrm{STA}_{x'y'}(\tau')$$

Stimulus autocorrelation

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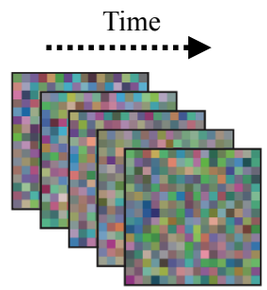
Linear-Gaussian model

s:



Completely random

$$w_{xy}(\tau) = \sum_{x'y'\tau'} \boxed{C^{-1}_{xy,x'y'}(\tau,\tau')}\, \mathrm{STA}_{x'y'}(\tau')$$

Stimulus autocorrelation

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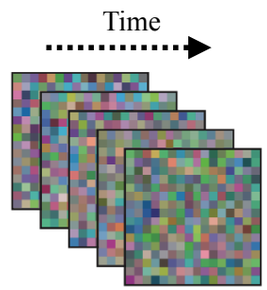
Linear-Gaussian model

If intensities are **random…**

s:



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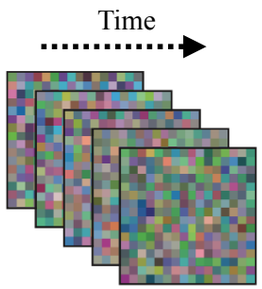
Linear-Gaussian model

If intensities are **random…**

$C_{xy,x'y'}(\tau', \tau) = \text{identity}$

s:



$$w_{xy}(\tau) = \sum_{x'y'\tau'} \boxed{C^{-1}_{xy,x'y'}(\tau, \tau')} \, \mathrm{STA}_{x'y'}(\tau')$$

Stimulus autocorrelation

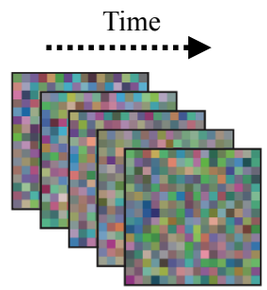
$$C_{xy,x'y'}(\tau', \tau) \equiv \frac{1}{T} \sum_{t}^{T} \tilde{S}_{xy}(t - \tau') \, \tilde{S}_{x'y'}(t - \tau)$$

Linear-Gaussian model

If intensities are **random…**

$C_{xy,x'y'}(\tau',\tau) = \text{identity} \Rightarrow w_{x,y}(\tau) = \text{STA}_{x,y}(\tau)$

s:



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Linear-Gaussian model

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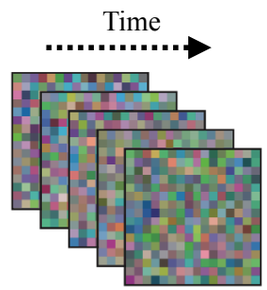
s:



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Linear-Gaussian model

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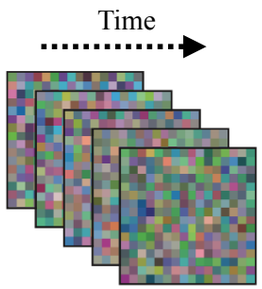

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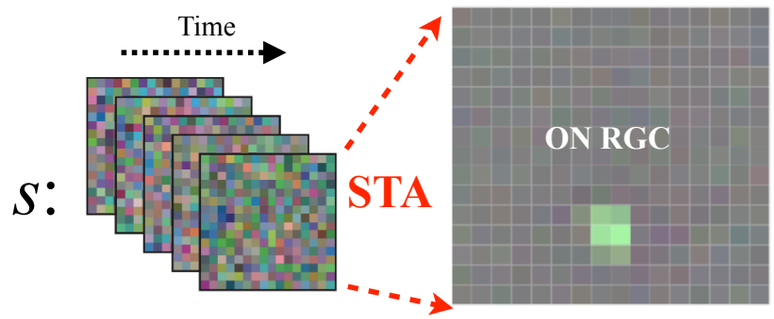

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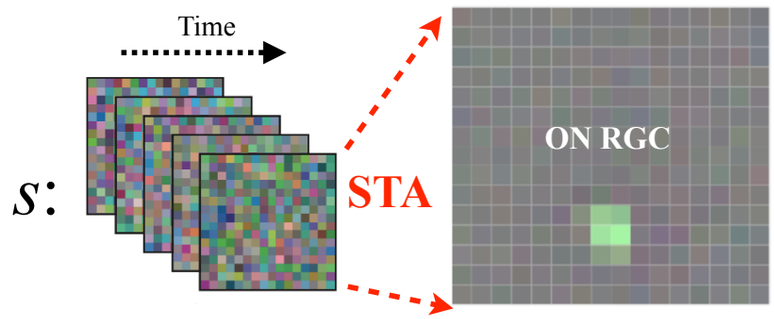

Example of STA
for an ON Retinal
Ganglion Cell (RGC)

Linear-Gaussian model

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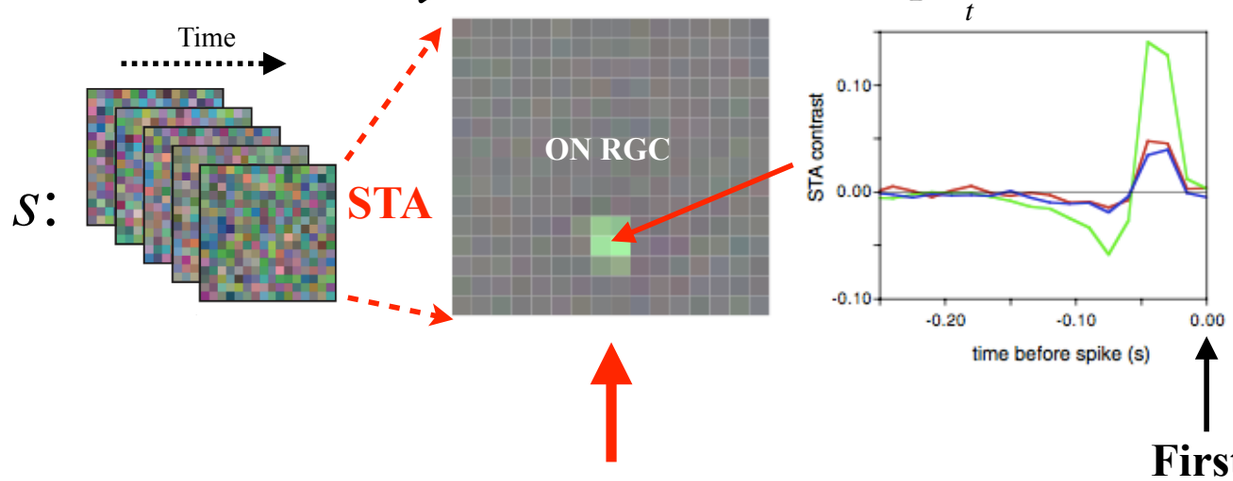

Example of STA
for an ON Retinal
Ganglion Cell (RGC)

First spike

Linear-Gaussian model

If intensities are **random…**

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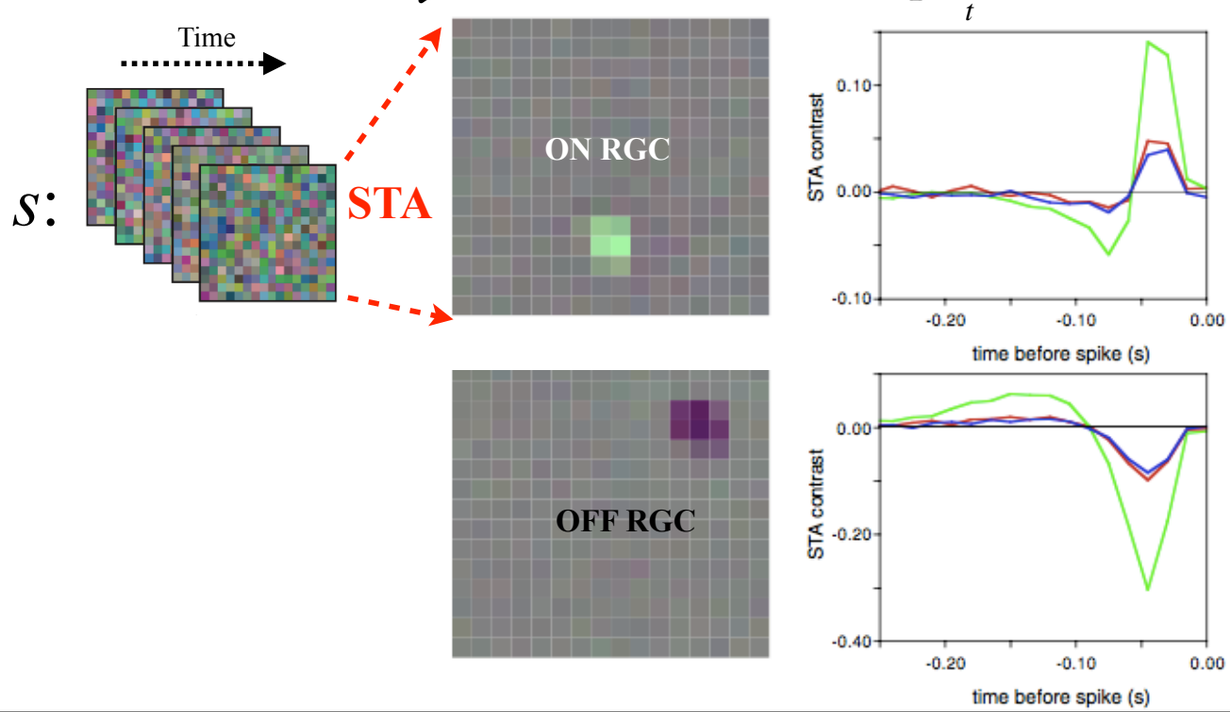
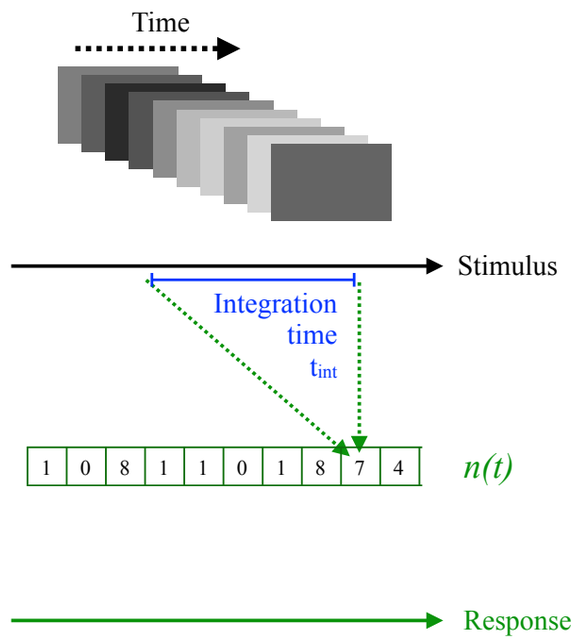


Linear-Gaussian model

If stimulus is **full field…**

$$f(t) = \boxed{w} * \tilde{S}(t) + b = \sum_{\tau}^{t_{int}} w(\tau)\ \tilde{S}(t-\tau) + b$$

$$\mathrm{MSE} = \frac{1}{2} \sum_{t} \left(f(t) - n(t) \right)^2$$



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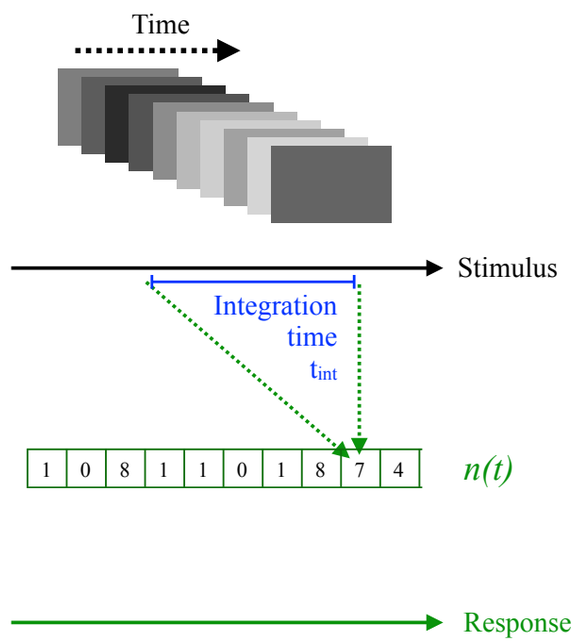

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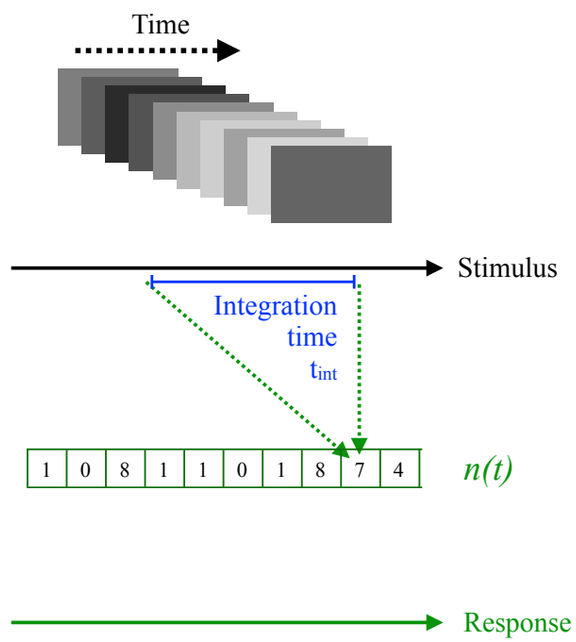

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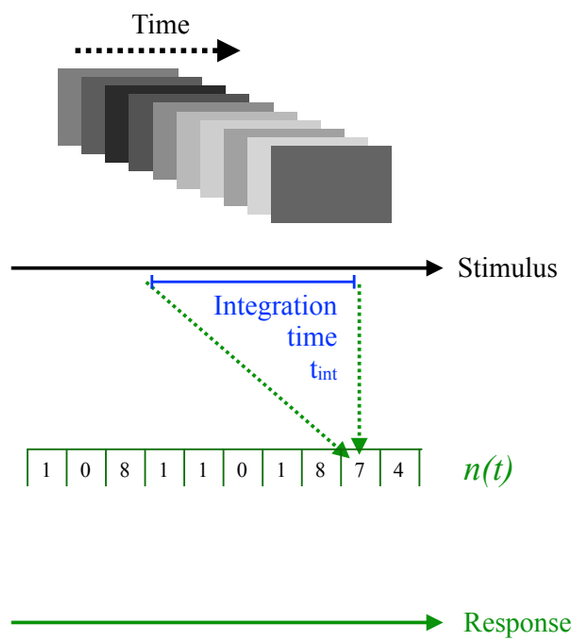

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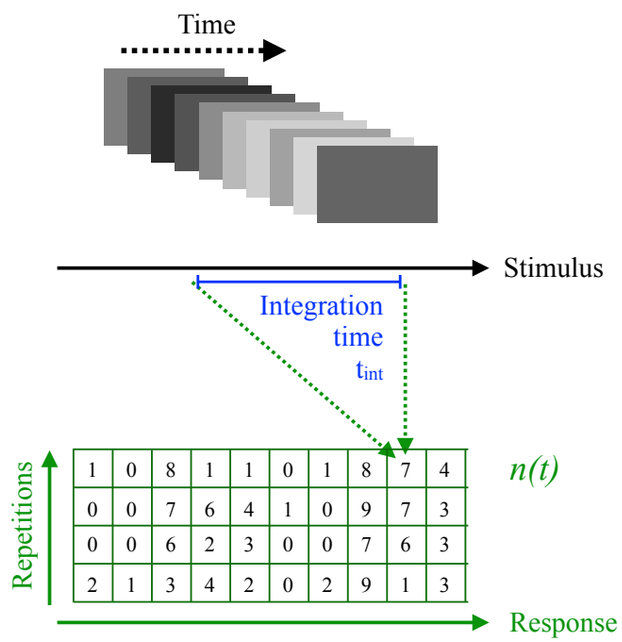

1	0	8	1	1	0	1	8	7	4
0	0	7	6	4	1	0	9	7	3
0	0	6	2	3	0	0	7	6	3
2	1	3	4	2	0	2	9	1	3

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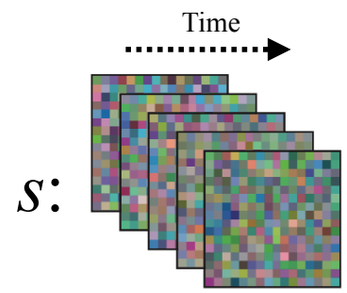
$$= \frac{1}{T}\sum_{t}\left(\frac{1}{R}\sum_{r} n^r(t)\right)\ \tilde{S}(t-\tau)$$

Repetitions

Linear-Gaussian model

If stimulus is **random…**



$$\boxed{w_{xy}(\tau)} = \sum_{x'y'\tau'} \boxed{C^{-1}_{xy,x'y'}(\tau,\tau')}\,\boxed{\mathrm{STA}_{x'y'}(\tau')}$$

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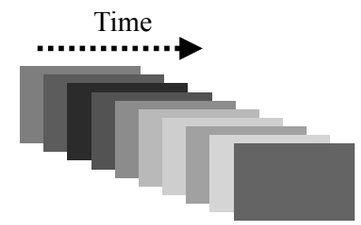

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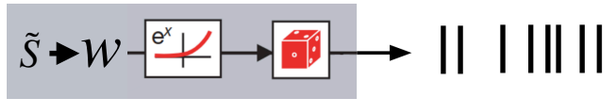
Linear-nonlinear Poisson model (LNP)

$$P_t(n) = Poisson\left(n \mid f(t)\right) = \frac{f(t)^n}{n!} e^{-f(t)}$$



$$f(t) = \mathrm{NL}\left(w * \tilde{S} + b\right) \rightarrow \exp\left(w * \tilde{S} + b\right)$$

Where NL(x) can be exp(x) or another nonlinear function

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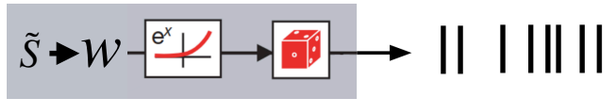

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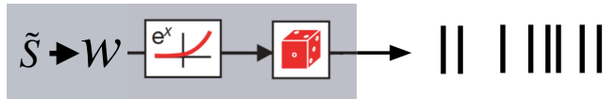

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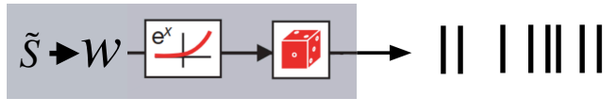

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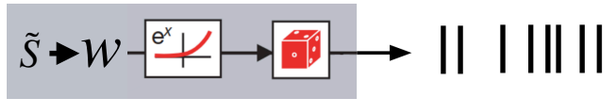

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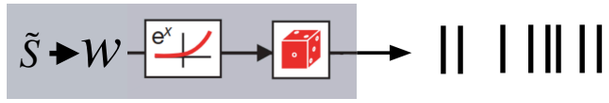

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$$\frac{\partial \ell(w,b)}{\partial b} = \frac{1}{T} \sum_t \left[\left(n(t) - f(t) \right) \right]$$

Can be solved by steepest gradient

LNP model

If stimulus is **full field…**

$$f(t) = \exp\left(w * \tilde{S} + b\right)$$

$$\ell(w,b) = \frac{1}{TR}\sum_{t,r}\left(n^r(t)\log f(t) - f(t)\right)$$

$$\frac{\partial \ell(w,b)}{\partial w(\tau)} = \frac{1}{TR}\sum_{t,r}\left[\left(n^r(t) - f(t)\right)\tilde{S}(t-\tau)\right]$$

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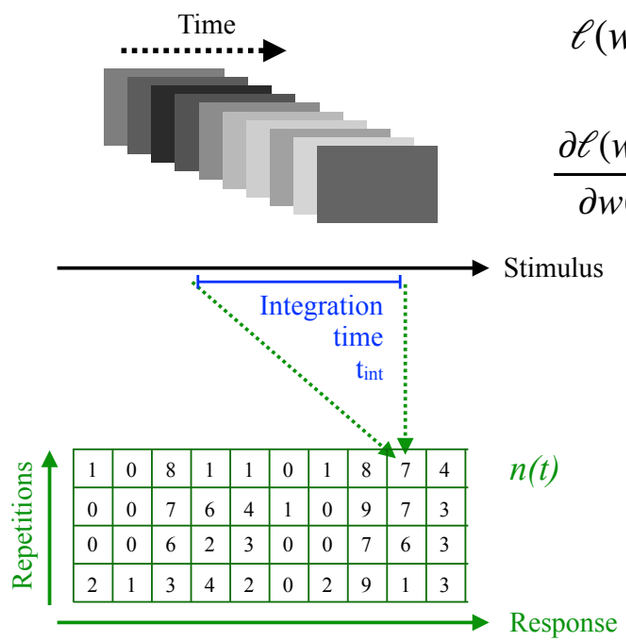

LNP model + L2-regularization

To have a smooth w, we should minimise $(\, w(\tau) - w(\tau + 1)\,)^2$

LNP model + L2-regularization

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$$\ell_\lambda(w, b) = \sum_t \left(n(t) \log f(t) - f(t) \right) - \frac{\lambda}{2} \sum_{\tau,\tau'} w(\tau)\, L(\tau, \tau')\, w(\tau')$$

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Laplacian: $$L(\tau,\tau') = \begin{pmatrix} 2 & -1 & & & & & & & \\ -1 & 4 & -1 & & & & & 0 & \\ & -1 & 4 & -1 & & & & & \\ & & -1 & \bullet & & & & & \\ & & & & \bullet & & & & \\ & & & & & \bullet & & & \\ & & & & & & -1 & & \\ & & & & & -1 & 4 & -1 & \\ & 0 & & & & & -1 & 4 & -1 \\ & & & & & & & -1 & 2 \end{pmatrix}$$

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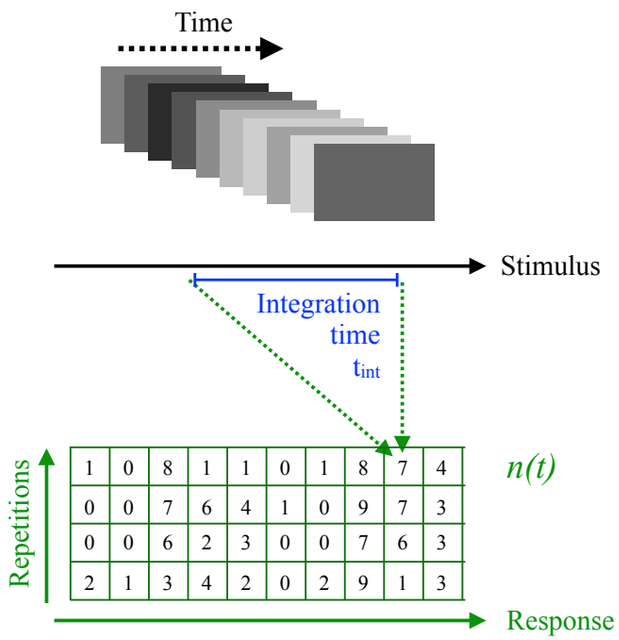
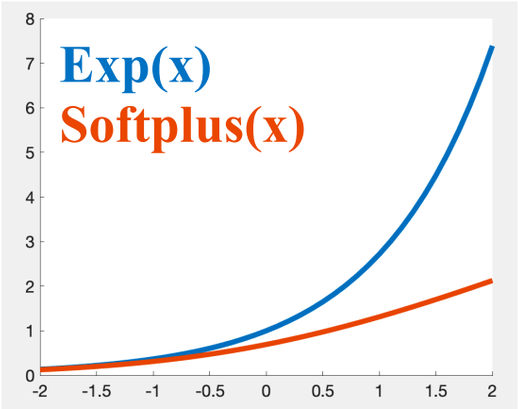
And the strength λ should be optimised over the validation set

LNP model

If stimulus is **full field…**

$$\mathrm{sp}(x) = \log\left(1 + e^{x}\right) \quad f(t) = \mathrm{sp}\left(w * \tilde{S} + b\right)$$





LNP model

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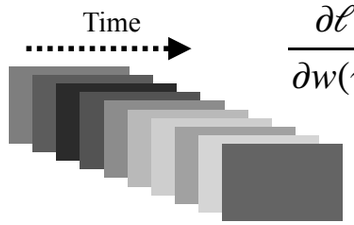
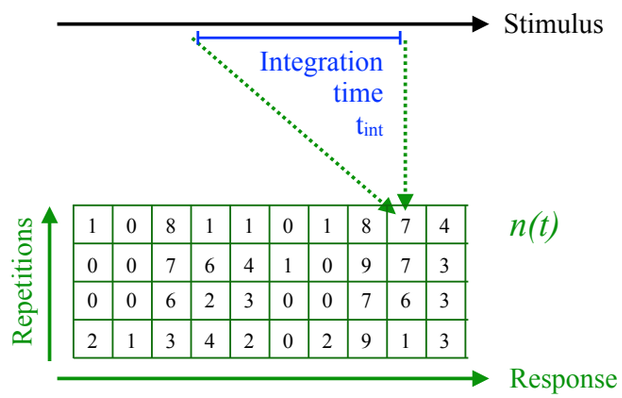
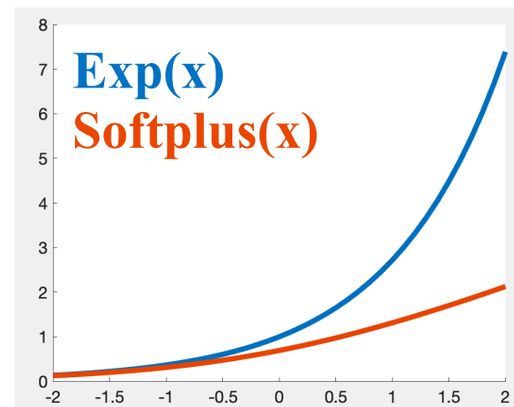
$$\mathrm{sp}(x) = \log\left(1 + e^{x}\right) \quad f(t) = \mathrm{sp}\left(w * \tilde{S} + b\right) \quad \ell = \frac{1}{TR}\sum_{t,r}\left(n^{r}(t)\log f(t) - f(t)\right)$$



$$\frac{\partial \ell}{\partial w(\tau)} = \frac{1}{TR}\sum_{t,r}\left[\left(\frac{n^{r}(t)}{f(t)} - 1\right)\mathrm{sp}'(w * \tilde{S} + b)\ \tilde{S}(t-\tau)\right]$$

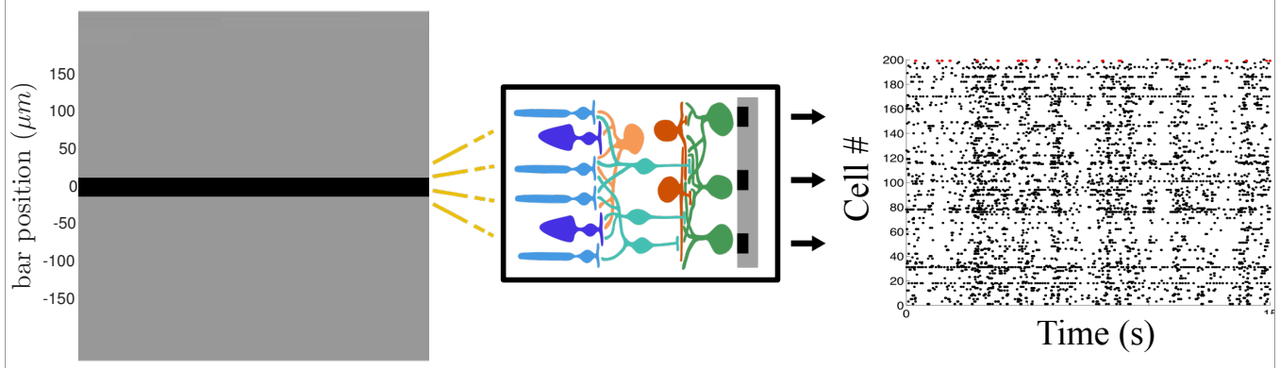
$$\frac{\partial \ell}{\partial b} = \frac{1}{TR}\sum_{t,r}\left[\left(\frac{n^{r}(t)}{f(t)} - 1\right)\mathrm{sp}'(w * \tilde{S} + b)\right]$$





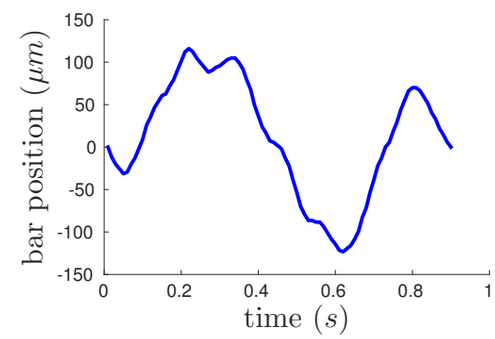
Predicting retinal light responses to complex stimuli

What if the stimulus is **more complex**?



Relevant: test the tracking of moving object

Simple: fully described by one dimensional trajectory



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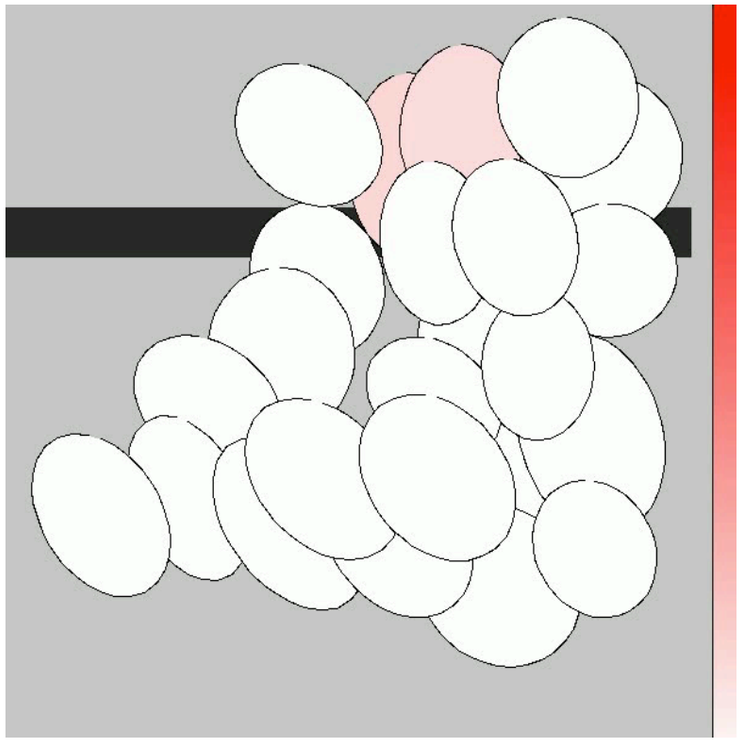

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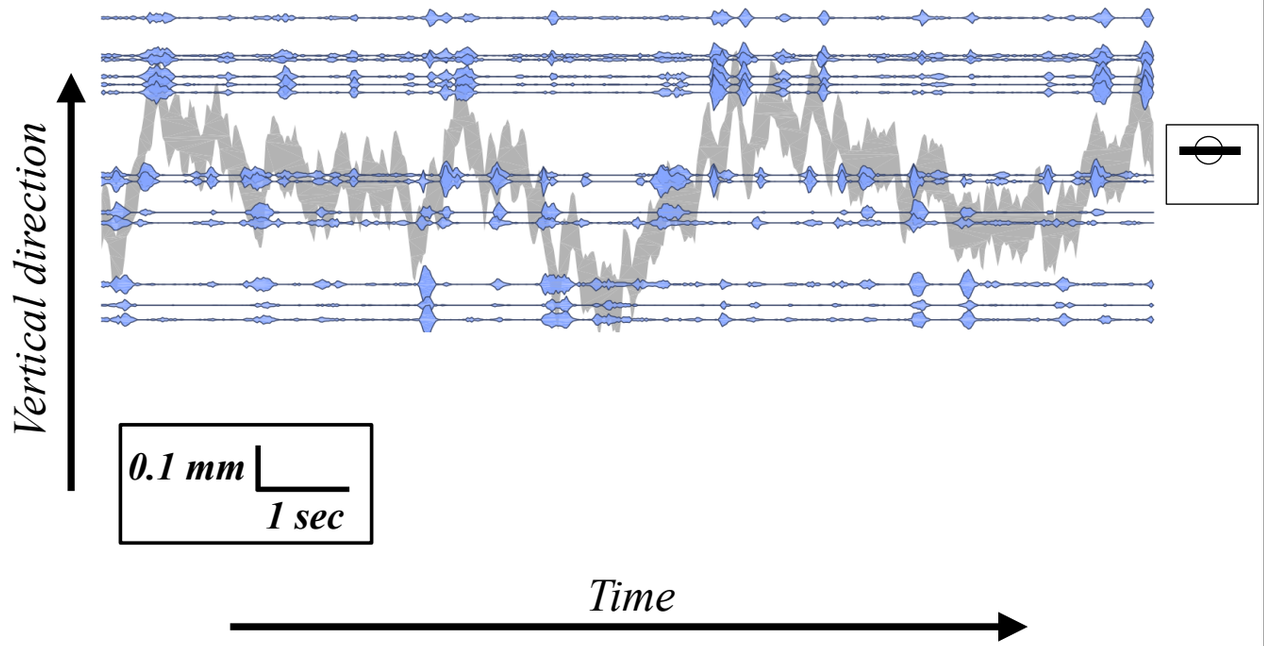
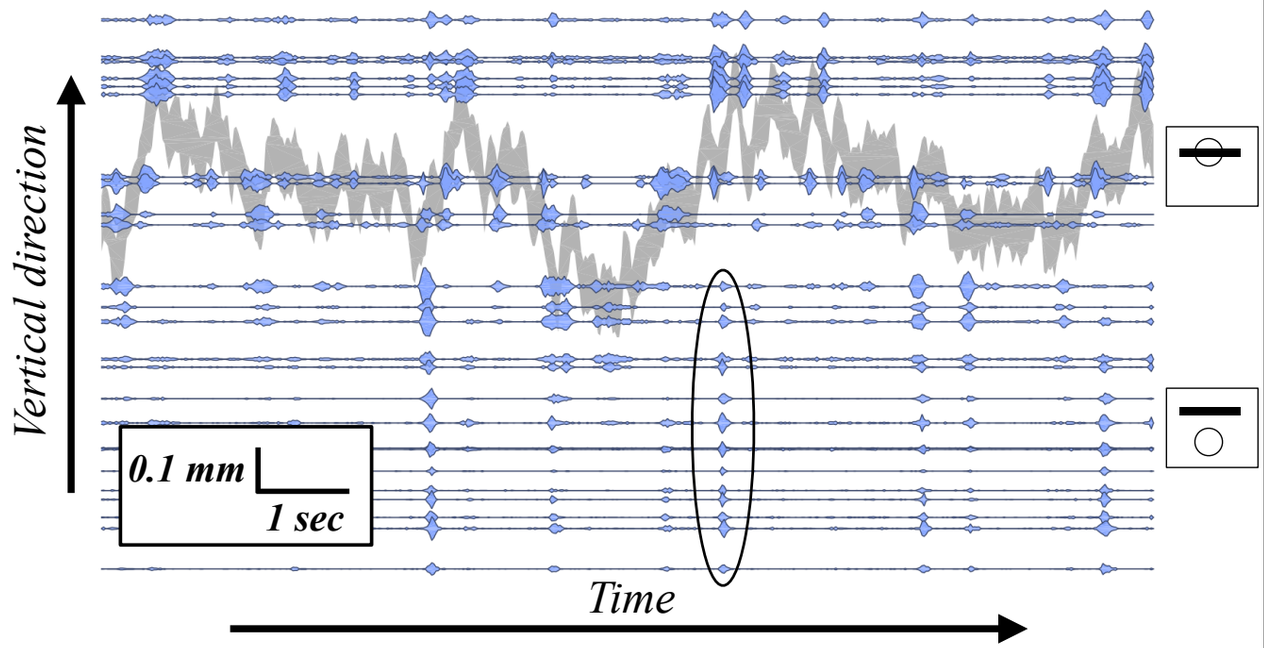


Predicting retinal light responses to complex stimuli



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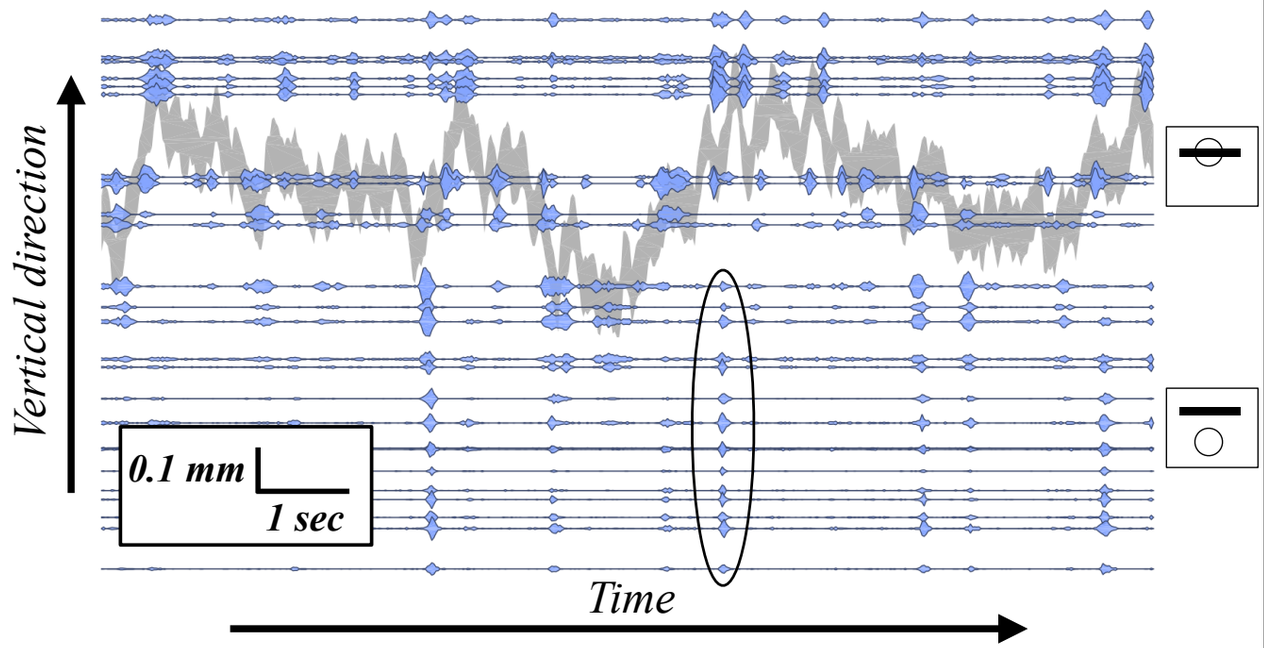
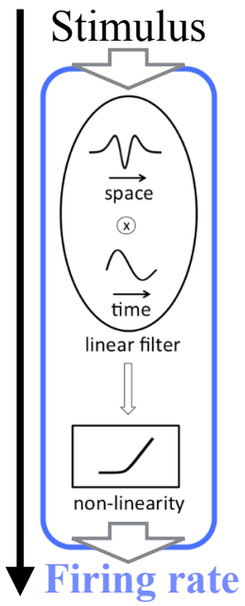
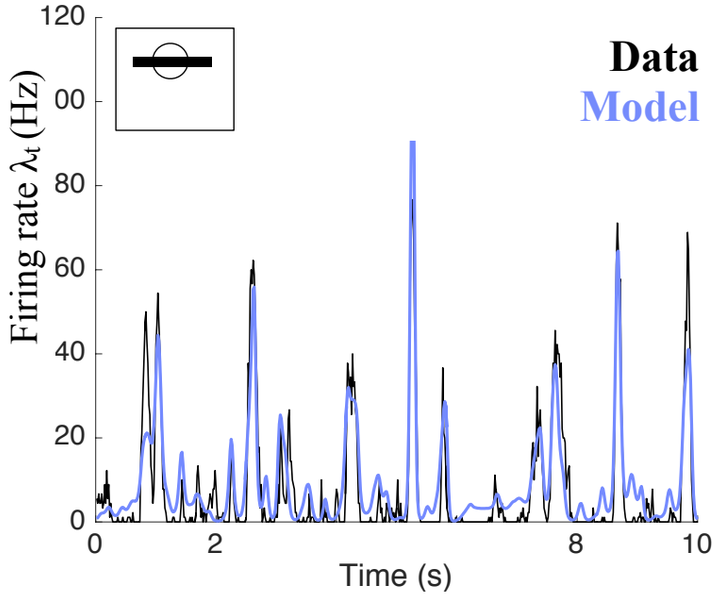


Same type of cell, but different computations!

Predicting retinal light responses to complex stimuli

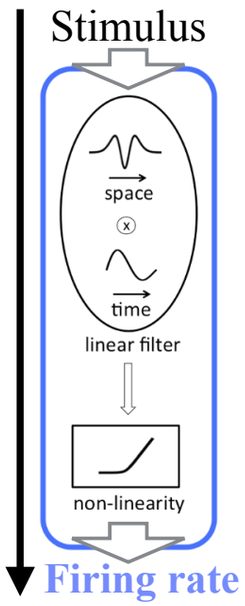
*A simple **LNP** model…*

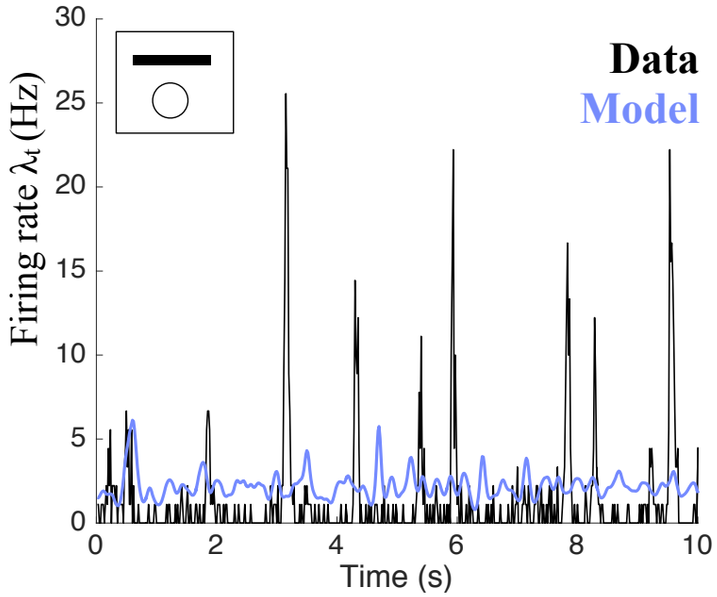




Predicting retinal light responses to complex stimuli

*A simple **LNP** model may not predict the response*





Predicting retinal light responses to complex stimuli

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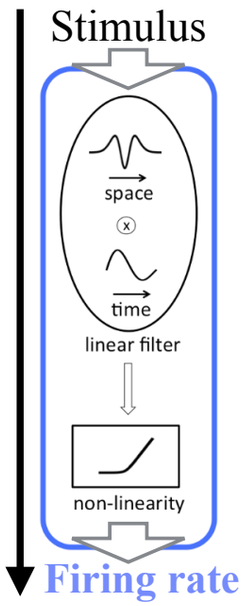
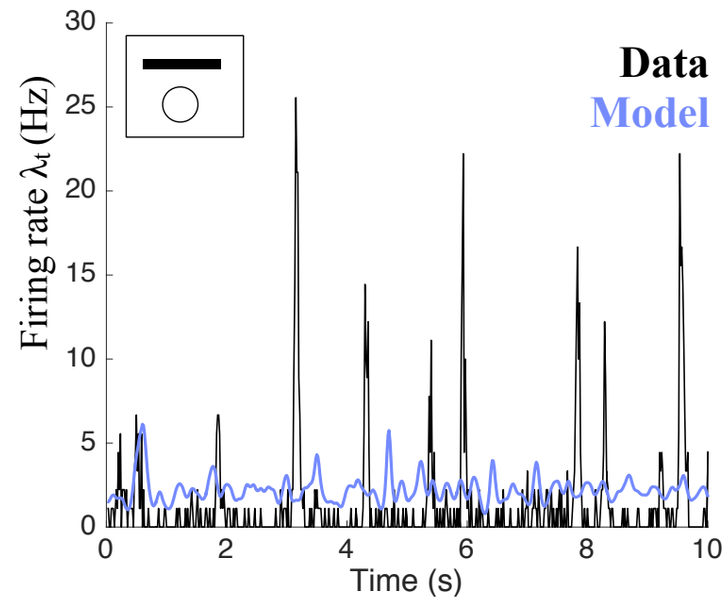
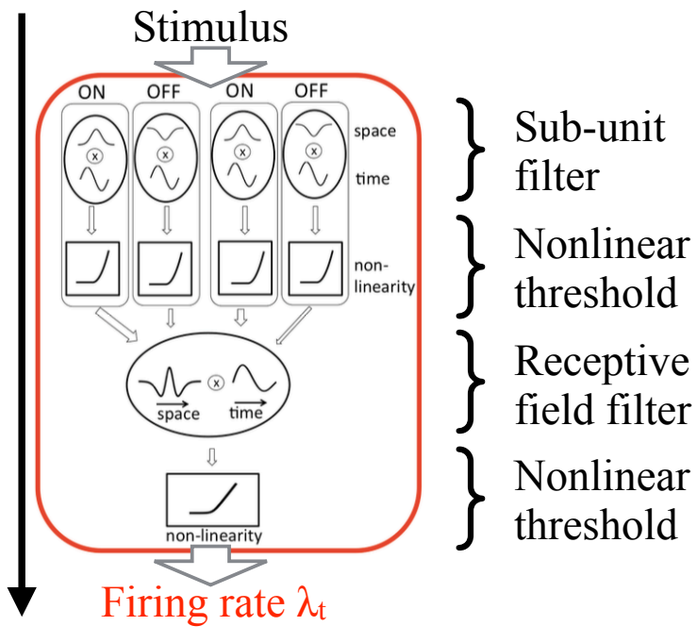




*A **LNP** model is not enough flexible!*

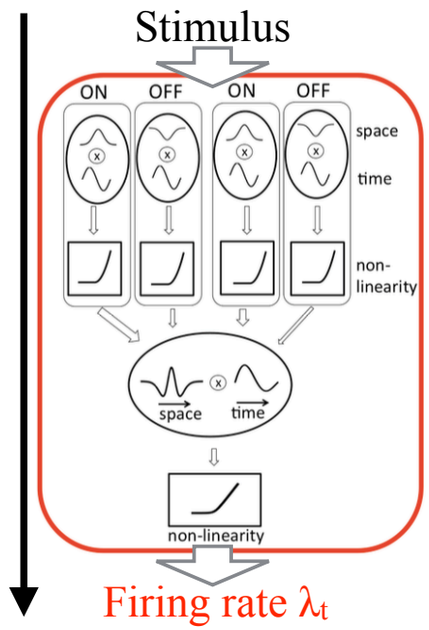
Predicting retinal light responses to complex stimuli

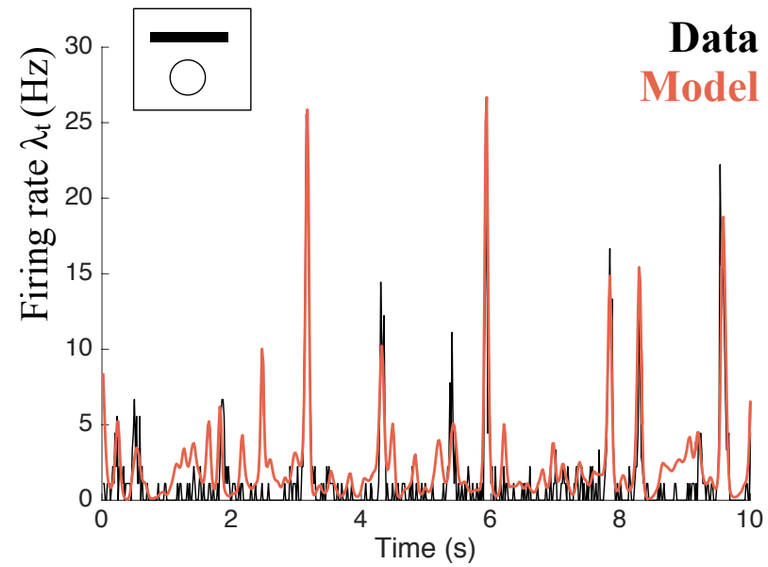
*An **highly non-linear** **LN²P** model…*



Predicting retinal light responses to complex stimuli

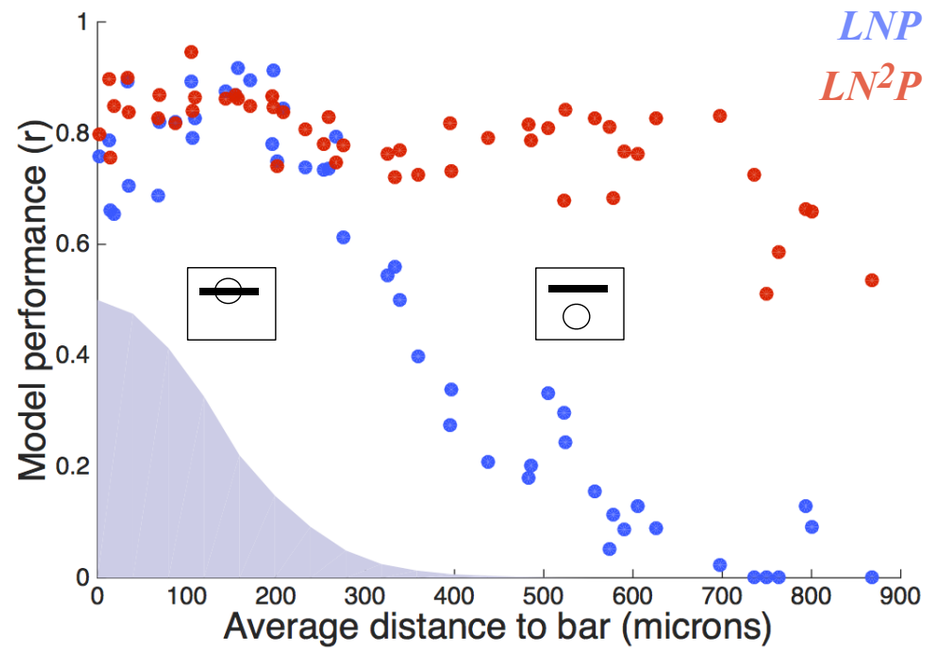
*An **highly non-linear** **LN²P** model predicts the response*





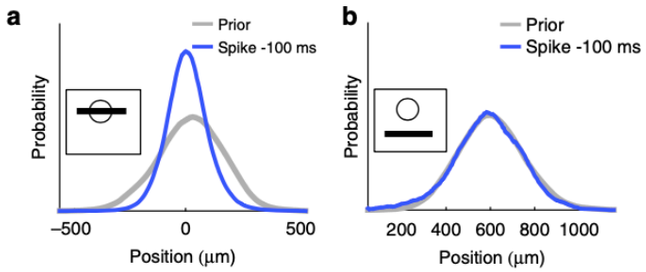
Predicting retinal light responses to complex stimuli

*An **highly non-linear** LN²P model predicts the response*



What are the neurons encoding?

How to quantify information carried by the spikes?



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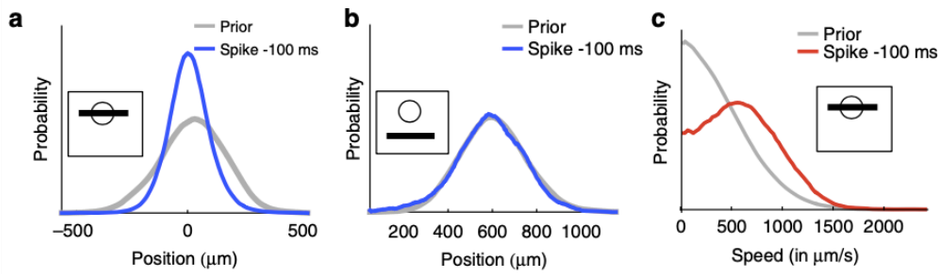

Same can be done for speed…

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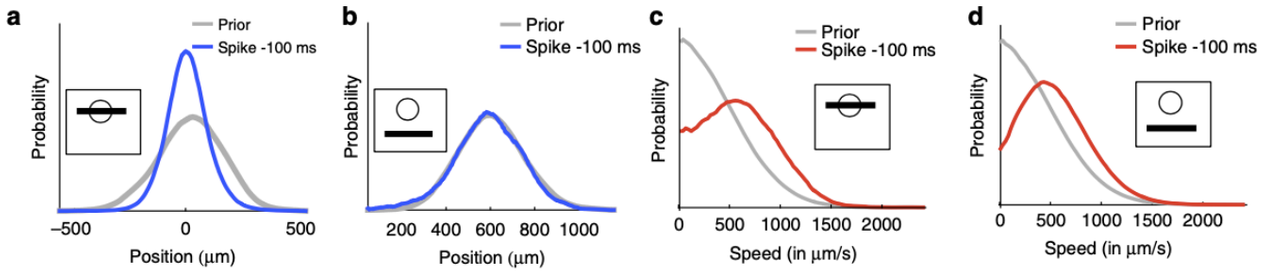

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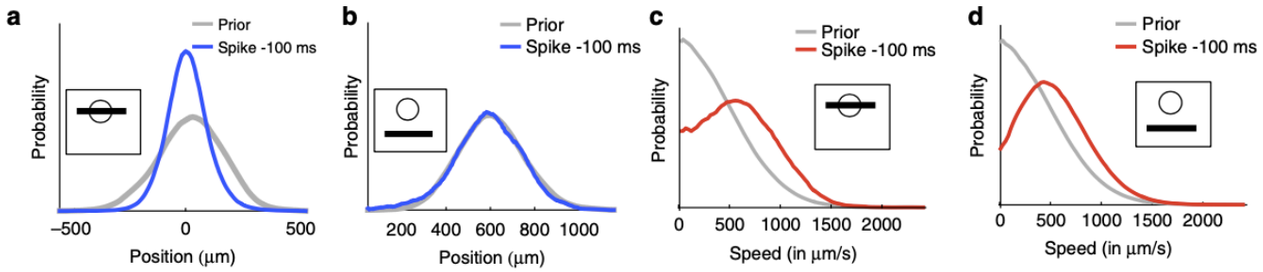

Mutual information between **pos** and **spikes**:

$$\mathcal{H}[\mathrm{pos}] - p(spike)\mathcal{H}[\,\mathrm{pos}\,|\,spike\,] - p(no\text{-}spike)\mathcal{H}[\,\mathrm{pos}\,|\,no\text{-}spike\,]$$

Same can be done for speed…

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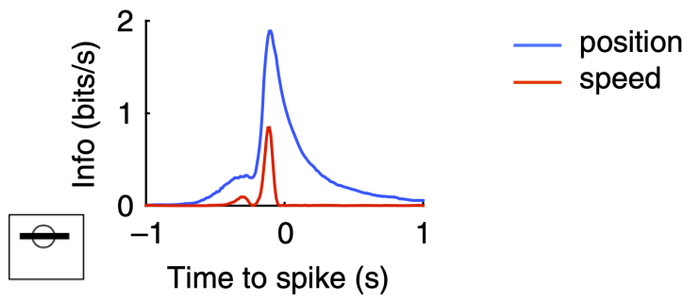

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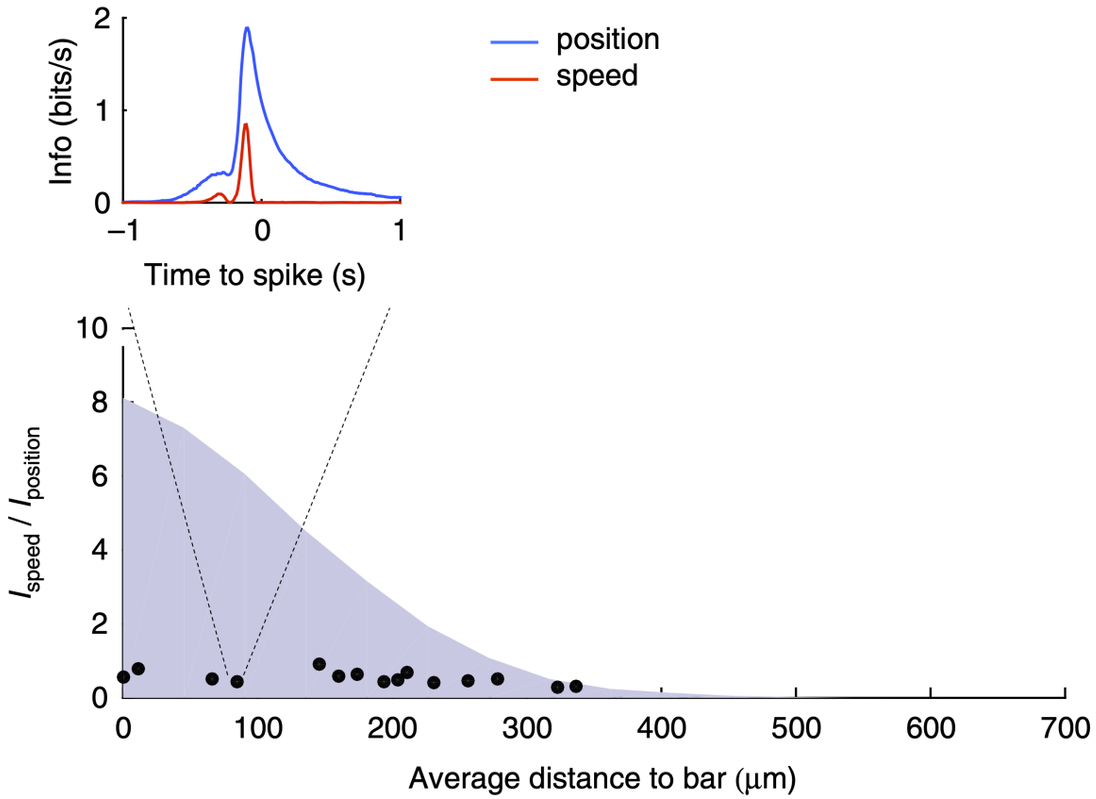

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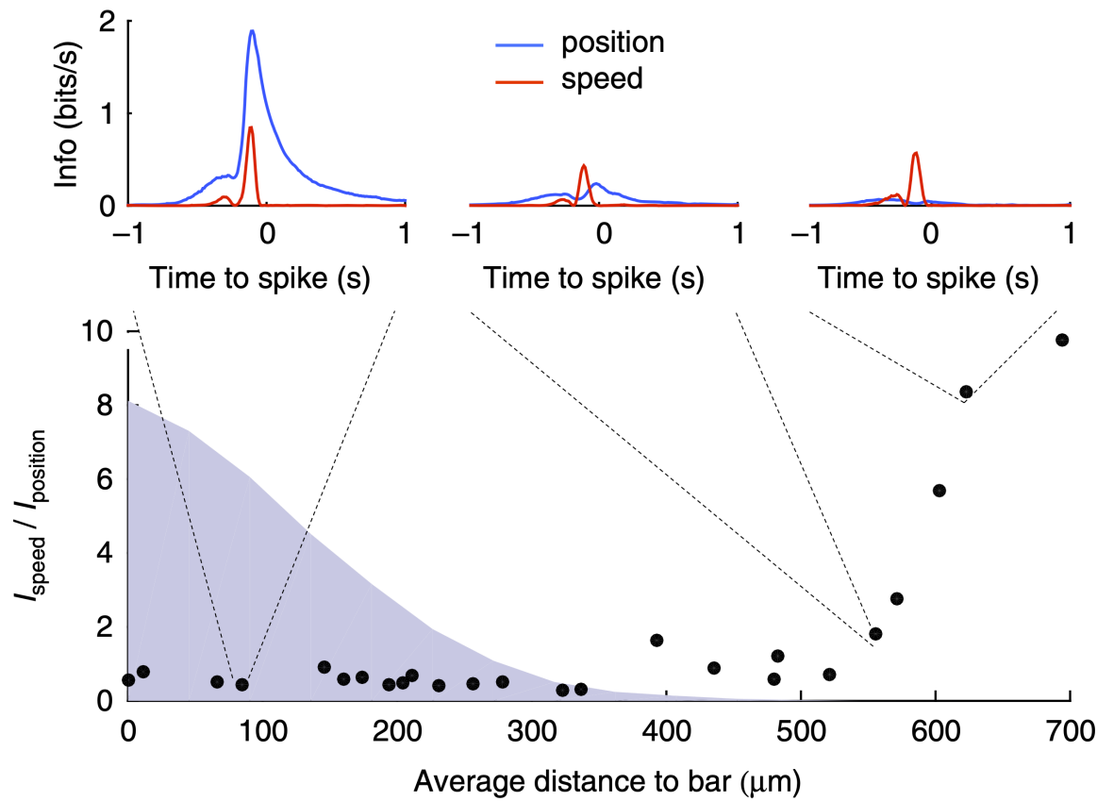
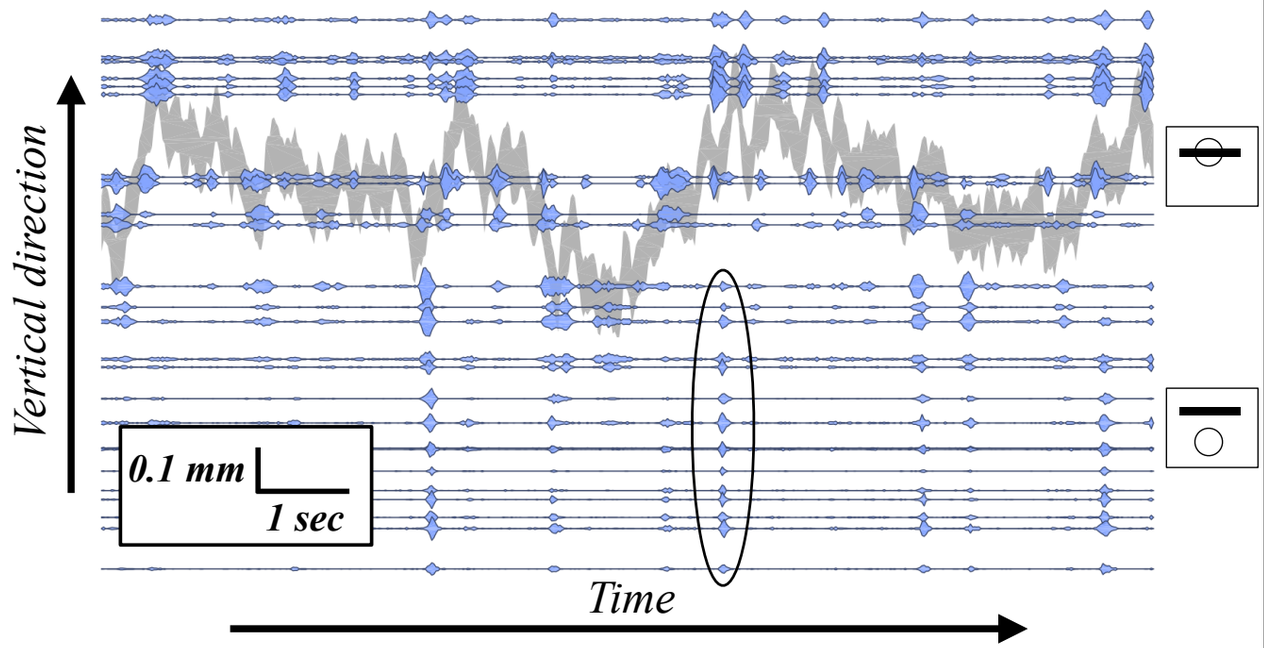


Predicting retinal light responses to complex stimuli



Same type of cell, but different computations!

Eyes are smarter than scientists believed

Steeve Laquitaine
& Ulisse Ferrari

laquitainesteeve@gmail.com

INSTITUT DE LA VISION PARIS

Material at **https://tinyurl.com/ensta2025**